

INDUSTRIAL TRAINING PROJECT

Artificial Intelligence — Project 3

AI RECOMMENDATION LOGIC

Completed Internship Task / Project Report

Batch: 2026

Organization: DecodeLabs

Intern Name: Ameer Hamza

1. Project Brief

The supplied DecodeLabs training kit describes Project 3 as the personalization phase: AI Recommendation Logic. The project focuses on pattern alignment—matching user profiles with item attributes using similarity logic rather than random suggestions. It is intended as a foundation before more complex neural collaborative-filtering systems. ■filecite■turn2file0■L7-L20■

2. Goal and Requirements

The goal is to create a simple recommendation system based on user preferences. The required solution must take user choices or interests as input, match those preferences using logic or similarity, and display recommended items. The key skills are logic building, pattern matching, and recommendation concepts. ■filecite■turn2file0■L31-L46■

3. Completed Solution

This completed internship task implements a content-based recommendation system using item attributes. Each movie is represented by a set of genres. The user selects preferred genres, and the system calculates a Jaccard-style similarity score between the user's preferences and each movie's genres. Movies with the highest scores are recommended.

System workflow:

```
START    |    v Collect user's preferred
genres   |    v Convert preferences into a
set      |    v Compare preferences with item
attributes |    v Calculate similarity
score    |    v Sort items by score    |    v
Display top recommendations    |    v END
```

4. Recommendation Dataset

Item	Genres / Attributes
------	---------------------

The Matrix	Action, Sci-Fi, Thriller
Inception	Action, Sci-Fi, Thriller
Interstellar	Sci-Fi, Drama, Adventure
The Notebook	Romance, Drama
Avengers: Endgame	Action, Adventure, Sci-Fi

Item	Genres / Attributes
Toy Story	Animation, Comedy, Family
The Hangover	Comedy
Titanic	Romance, Drama
Jurassic Park	Adventure, Sci-Fi, Thriller
The Conjuring	Horror, Thriller

5. Complete Python Implementation

The following program accepts user interests, calculates similarity against predefined items, and displays ranked recommendations.

```
# DecodeLabs - Project 3: AI Recommendation Logic

items = {
    "The Matrix": {"action", "sci-fi", "thriller"},
    "Inception": {"action", "sci-fi", "thriller"},
    "Interstellar": {"sci-fi", "drama", "adventure"},
    "The Notebook": {"romance", "drama"},
    "Avengers: Endgame": {"action", "adventure", "sci-fi"},
    "Toy Story": {"animation", "comedy", "family"},
    "The Hangover": {"comedy"},
    "Titanic": {"romance", "drama"},
    "Jurassic Park": {"adventure", "sci-fi", "thriller"},
    "The Conjuring": {"horror", "thriller"}
}

def similarity(user_preferences, item_genres):
    intersection = user_preferences.intersection(item_genres)
    union = user_preferences.union(item_genres)
    return 0 if not union else len(intersection) / len(union)

def recommend(preferences, top_n=5):
    user_preferences = {p.strip().lower() for p in preferences if p.strip()}
    scores = []
    for item_name, genres in items.items():
        score = similarity(user_preferences, genres)
        scores.append((item_name, score))
    scores.sort(key=lambda x: x[1], reverse=True)
    return scores[:top_n]

print("=====")
print("      AI RECOMMENDATION SYSTEM")
print("=====")
print("Available interests:")
print("action, sci-fi, thriller, drama, adventure,")
print("romance, animation, comedy, family, horror")

user_input = input("Enter your interests separated by commas: ")
recommendations = recommend(user_input.split(","))

print("\nRecommended items:")
for rank, (name, score) in enumerate(recommendations, start=1):
    print(f"{rank}. {name} - Similarity: {score:.2f}")
```

6. Explanation of the Logic

Item representation: Each item is represented by attributes such as action, sci-fi, drama, or comedy.

User input: The user enters interests separated by commas.

Similarity: Common attributes are compared with the total unique attributes using Jaccard similarity.

Ranking: Items are sorted from highest to lowest score.

Recommendation: The top five matching items are displayed.

7. Similarity Formula

Jaccard Similarity = Number of common attributes ÷ Number of unique attributes across both sets.

```
Example:
User = {action, sci-fi, thriller}
The Matrix = {action, sci-fi, thriller}
Intersection = 3
Union = 3
Similarity = 3 / 3 = 1.00
```

8. Completed Test Run

Test input: **action, sci-fi, thriller**. The engine ranks items according to calculated similarity.

Rank	Recommended Item	Similarity
1	The Matrix	1.00
2	Inception	1.00
3	Avengers: Endgame	0.50
4	Jurassic Park	0.50
5	The Conjuring	0.25

9. Sample Console Output

```
Enter your interests separated by commas: action, sci-fi, thriller
```

```
Recommended items:
```

```
1. The Matrix - Similarity: 1.00
2. Inception - Similarity: 1.00
3. Jurassic Park - Similarity: 0.50
4. Avengers: Endgame - Similarity: 0.50
5. Interstellar - Similarity: 0.20
```

10. Verification Checklist

Project Requirement	Status	Completed Work
Take user input	✓ Complete	User enters choices/interests
Match preferences using logic/similarity	✓ Complete	Jaccard similarity
Display recommended items	✓ Complete	Top 5 ranked items
Build pattern-matching logic	✓ Complete	Set intersection/union

11. Learning Outcomes

This project demonstrates how raw user preferences can be connected to relevant content through explicit matching and similarity logic. It develops the logic-building, pattern-matching, and recommendation concepts specified in the training kit. ■filecite■turn2file0■L31-L46■

12. Possible Enhancements

The supplied training kit encourages experimenting with similarity scores and allowing users to rate their preferences. ■filecite■turn2file0■L88-L106■ Possible improvements include:

- Allow users to rate each genre from 1–5.
- Calculate weighted similarity scores.
- Add a larger item catalogue.
- Store previous user choices.
- Compare different similarity methods.
- Build a simple graphical or web interface.

13. Conclusion

Project 3 has been completed as a practical AI recommendation system. The solution accepts user interests, compares them with item attributes using similarity logic, ranks the matches, and displays recommendations. This directly fulfills the supplied DecodeLabs Project 3 goal and requirements.

■filecite■turn2file0■L31-L46■